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Gesture controlled Virtual Mouse using Computer vision

Abstract

The Virtual Mouse & Quantum Assistant is an AI-based system that lets users control their computer without a physical mouse or keyboard. It uses hand tracking to enable actions like moving the cursor, clicking, scrolling, and adjusting volume or brightness through gestures. It also includes Quantum, an AI assistant that supports voice and text commands, features conversational memory, integrates with language models, and provides a chat interface.

Introduction

Traditional computer interaction depends on physical devices like keyboards and mice, which can be limiting - especially for users with mobility challenges - and less natural overall. This project introduces a touchless alternative using computer vision and machine learning on standard hardware.

It features two modes:

- Gesture Mode, where a webcam-based virtual mouse enables cursor control, clicks, scrolling, and system adjustments through hand gestures.
- Quantum Assistant Mode, a voice and text-based chatbot that executes a wide range of system commands using fuzzy matching and an LLM-powered backend.

Objectives

- To develop a real-time, touchless mouse control system using hand gesture recognition via webcam, eliminating the need for physical input devices.
- To implement voice and text-driven AI assistant capable of dispatching 31 categories of system commands with fuzzy matching for improved accuracy.
- To integrate a cascading LLM backend with conversation memory for intelligent, context-aware responses to unrecognized commands.
- To enable custom gesture training, allowing users to define and map personal gestures to system actions such as hotkeys, app launches, or text input.
- To provide an accessible and configurable interface through a browser-based chat UI with real-time settings for gesture sensitivity, cursor speed, and pinch threshold.

Methodology / System Design

The Virtual Mouse & Quantum Assistant follows a modular architecture separating gesture processing, assistant logic, and LLM integration for efficient and maintainable system design.

1.Gesture Recognition Module:
The webcam captures live frames processed by MediaPipe Hands to extract 21 hand landmarks per frame. The HandRecog class classifies these landmarks into gesture enums using finger state detection. Gestures are stabilized over 4+ consecutive frames to eliminate jitter.

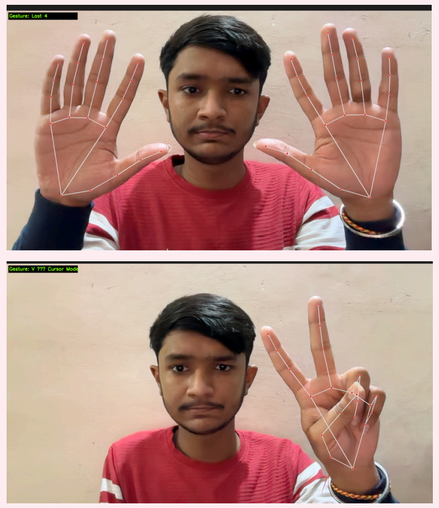

Figure 1 :Gesture recognition and cursor movement

2. System Control Module:
The Controller class maps recognized gestures to system actions - cursor movement, left/right click, double click, drag, scroll, volume, and brightness control. PyAutoGUI handles cross-platform mouse actions, with Windows-specific audio/brightness libraries loaded conditionally.

Figure 2 :Gesture-to-action mapping interface

3. Quantum Assistant - Command Dispatcher:
User voice or text input passes through fuzzy matching to correct typos, then routes to the respond() dispatcher covering 31 command categories (app control, browser tabs, file search, music, timers, unit conversion, Wikipedia, and more).

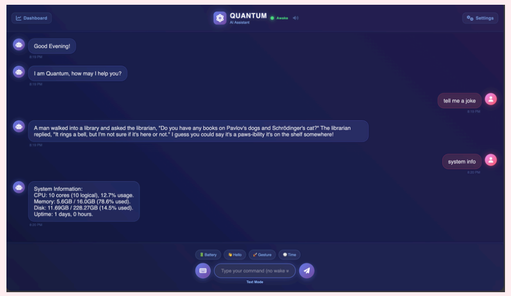

Figure 3: Quantum chat UI

4 LLM Integration & Conversation Memory:
Unrecognized commands are routed to a LangChain agent backed by Gemini 2.5 Flash, Groq LLaMA-3.3-70B, or Ollama. The agent retains the last 8 conversation turns as memory and has access to 9 action tools (web search, file search, weather, calculator, screenshot, etc.).

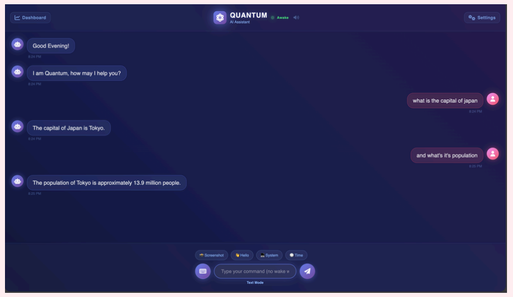

Figure 4:LLM-powered response with conversation context

5. Live Dashboard & Settings Panel:
A browser-based Eel UI displays a real-time dashboard with gesture heatmap, command history, and CPU/RAM stats. The settings panel allows tuning gesture sensitivity, cursor speed, pinch threshold, and scroll speed - changes apply on next gesture session start.

Figure 5 : Live dashboard and settings panel

Conclusion

The Virtual Mouse & Quantum Assistant showcases an innovative approach to human-computer interaction by combining gesture-based control with an AI voice assistant. Using technologies like MediaPipe, OpenCV, and LLMs, it removes the need for physical input devices while enabling smart, context-aware commands. The system is accessible, cross-platform, and scalable, making it well-suited for modern, touchless computing environments.

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